

IIT Foundation Module for Class VI

Chemistry

- Theme 1: Matter and its Composition
 1. Definition of matter
 2. Matter has mass and occupies space – explanation
 3. Composition of matter – brief introduction

- Theme 2: Physical and Chemical Changes
 1. Physical and chemical changes
 2. Chemical change – formation of a new product with new properties
 3. Differentiating between physical and chemical change
 4. Classification as physical and chemical change
 5. Types of change involved when there is a change of state of matter
 6. Types of change involved when there is a change in energy

- Theme 3: Elements, Compounds and Mixtures (experimental techniques)
 1. Identification of elements and compounds from representation of their symbols and formulae
 2. Mixtures and compounds: difference between mixtures and compounds on the basis of the chemical composition of constituents
 3. Recall that a mixture is formed when two or more substances are mixed in any proportion such that their particles are in intimate contact with one another without undergoing a chemical change
 4. Types of mixtures
 5. Separation techniques
 6. Examine the principle behind each separation technique
 7. Chromatography as a separation technique; Paper chromatography

- Theme 4: Atomic Structure
 1. An atom is the smallest particle of an element
 2. It is not capable of independent existence

3. The properties of an element depend upon the atoms constituting it
4. A molecule is the smallest particle of an element or compound, capable of independent existence. It consists of one or more atoms of the same or different elements
5. A radical is a single atom of an element or a group of atoms of different elements behaving as a single charged unit
6. Atomicity (number of atoms in an entity) of elements and compounds – monoatomic, diatomic, triatomic, polyatomic
7. Associate the first 20 elements in the periodic table with their names and symbols
8. Valency is the combining capacity of an element or the number of hydrogen atoms with which it combines or replaces

- Theme 5: Language of Chemistry

1. A chemical reaction may take place when two or more reactants come in contact with one another and transfer of energy takes place
2. Characteristics of occurrence of a chemical reaction:
 - Change of colour
 - Change of state
 - Change of smell
 - Evolution of gas
 - Precipitate formed
 - Heat evolved / released
3. Chemical Equations:
 - Writing word equations for chemical reactions and emphasizing observational skills and the names of products formed
 - Some examples of word equations for practice

- Theme 6: Metals and Non-Metals

1. Properties
2. Distinguish between metals and non-metals with the general properties (lustre, conduction of electricity, heat, malleability, ductility, sonority, melting point, boiling point, density, strength)
3. Classification of elements as metals and non-metals
4. Corrosion of iron (rusting); ways to prevent rusting (oiling, painting, chrome plating, galvanization, tinning) and avoiding contact with air and water vapour
5. Uses of certain metals (iron, gold, copper, aluminium, zinc, lead, magnesium)

6. Metalloids: elements that show the properties of both metals and non-metals (e.g., silicon, germanium, tungsten, antimony) and their uses

- Theme 7: Air and Atmosphere

1. Air as a mixture of gases
2. Composition of air and uses of its components
3. Oxygen is needed for combustion
4. Mass change during burning (burning of magnesium and candle)
5. Word equations for reactions of metals and non-metals (S, C, P, Na, K, Ca, Mg) with O
6. Products formed in acid rain; effects of acid rain
7. Air quality
8. Study the properties of oxygen (physical properties to include colour and odour)
9. Distinguish between